Pharmacist-Led Group Medical Appointments for the Management of Type 2 Diabetes with Comorbid Depression in Older Adults

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BACKGROUND: Depression is associated with poor glycemic control, increased number of microvascular and macrovascular complications, functional impairment, mortality, and 4.5 times higher total health care costs in patients with diabetes. Shared medical appointments (SMAs) may be an effective method to attain national guideline recommendations for glycemic control in diabetes for patients with depression through peer support, counseling, problem solving, and improved access to care.

OBJECTIVE: To test the efficacy as assessed by attainment of a hemoglobin A_{1c} (A1C) <7% of pharmacist-led group SMA visits, Veterans Affairs Multidisciplinary Education in Diabetes and Intervention for Cardiac Risk Reduction in Depression (VA-MEDIC-D), in patients with type 2 diabetes mellitus.

METHODS: This was a randomized controlled trial of VA-MEDIC-D added to standard care versus standard care alone in depressed patients with diabetes with A1C >6.5%. VA-MEDIC-D consisted of 4 once-weekly, 2-hour sessions followed by 5 monthly 90-minute group sessions. Each SMA session consisted of multidisciplinary education and pharmacist-led behavioral and pharmacologic interventions for diabetes, lipids, smoking, and blood pressure. No pharmacologic interventions for depression were provided. The change in the proportion of participants who achieved an A1C <7% at 6 months was compared.

RESULTS: Compared to standard care (n = 44), a lower proportion of patients in VA-MEDIC-D (n = 44) had systolic blood pressure (SBP) <130 mm Hg at baseline, but were similar in other cardiovascular risk factors and psychiatric comorbidity. The change in the proportion of participants achieving an A1C <7% was greater in the VA-MEDIC-D arm than in the standard care arm (29.6% vs 11.9%), with odds ratio 3.6 (95% CI 1.1 to 12.3). VA-MEDIC-D participants also achieved significant reductions in SBP, low-density lipoprotein cholesterol, and non–high-density lipoprotein (HDL) cholesterol from baseline, whereas significant reductions were attained only in non–HDL cholesterol with standard care. There was no significant change in depressive symptoms for either arm.

CONCLUSIONS: Pharmacist-led group SMA visits are efficacious in attainment of glycemic control in patients with diabetes and depression without change in depression symptoms.

KEY WORDS: depression, diabetes, pharmacy practice.

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The prevalence of comorbid depression in patients with diabetes is 30%, nearly twice the rate of the general population.[1] For patients with diabetes, depression is associated with poor glycemic control, increased number of microvascular and macrovascular complications, and a 2.3-fold increase in mortality.[2-8] The combined effect of having diabetes and depression has been recognized as particularly problematic, as each condition negatively influences the other and results in decreased adherence to dietary recommendations and oral medications, poorer health-related outcomes, and increased health care utilization and costs.[2,9-12]

Previous reports have shown that the presence of depression attenuates the effects of diabetes interventions, while treating depression alone fails to improve diabetes outcomes.[2,5,11] Possible explanations are that depressive symptoms are associated with social withdrawal, poor physical and mental functioning, and low motivation, skill level, and self-confidence secondary to poor disease knowledge and perception of limited benefits of behavioral performance,[13-15] all of which limit social support and self-care, the basis of diabetes management. Patients with chronic comorbid physical and mental conditions, especially depression, are also less

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likely to seek care and are more likely to receive substandard preventive diabetes care, possibly because of social stigma or competing demands for primary care physicians' time.[2,5]

Shared medical appointments (SMAs) may be an effective method to address care barriers in diabetes for those with depression through peer support, counseling, problem solving, empowerment, and improvement of access to care.[16,17] Weinger defined SMAs as group visits in which several patients meet with the same provider(s) at the same time.[16] SMAs that are efficacious have relied on intensive physician involvement working in collaboration with a multidisciplinary health care team to perform aggressive pharmacologic interventions.[18-20] Using constructs from the chronic care model, we propose a division of labor with the use of clinical pharmacists to deliver pharmacologic interventions and self-management support for the treatment of diabetes in depression.[21] Clinical pharmacists are able to provide these interventions through collaborative practice agreements. These pharmacotherapeutic services are potentially reimbursable.[22] We have performed studies to show that a pharmacist-led SMA model (Veterans Affairs Multidisciplinary Education in Diabetes and Intervention for Cardiac Risk Reduction [VA-MEDIC]) using disease self-management education and aggressive medication titration by the clinical pharmacist is effective in improving hyperglycemia and cardiovascular risk factors in a general diabetes population.[23,24] However, it is unknown how well our pharmacist-led SMA would work in a historically treatment-resistant depressed population with diabetes and how durable this intervention would be on diabetes risk factor control for 6 months.[5,25] This study sought to determine whether SMAs are feasible for the treatment of diabetes in patients with depression and to evaluate whether these can be efficacious when led by nonphysician professionals such as clinical pharmacists with prescribing authority.

Methods

DESIGN

We performed a randomized controlled trial to test the feasibility of a multidisciplinary education and pharmacist-led intensive behavioral and pharmacologic group intervention in the treatment of diabetes and cardiovascular risk factors in patients with comorbid depression (VA-MEDIC-D). The study compared VA-MEDIC-D added to standard care versus standard care alone in the treatment of hyperglycemia, hypertension, hyperlipidemia, and cigarette smoking in patients with diabetes and depression. Participants were assigned to the VA-MEDIC-D arm or standard care arm using simple coin toss randomization. The Institutional Review Board at the Providence Veterans Affairs Medical Center (VAMC) approved the protocol, and all study procedures were conducted in accordance with the ethical standards of the Helsinki Declaration of 1975. Enrollment for this study began December 2006 and ended November 2008 (Figure 1).

POPULATION

Eligible patients were identified by a combination of review of the Providence VAMC electronic medical record system and referral by primary care providers. The study population included all veterans with type 1 and type 2 diabetes (T2DM) with hemoglobin A_{1c} (A1C) >6.5% within the last 6 months and concomitant depression, as defined by ICD-9 codes 311, 296.2, and 296.3, who were willing to participate and discuss their diabetes and cardiovascular risk factors in a group setting and able to provide written informed consent (Appendix I). Since enrollment began in 2006, our enrollment criteria were developed to be consistent with the 2006 American Diabetes Association (ADA) treatment guidelines for glycemic control, which had indicated that the A1C goal for individuals should be as close to normal (<6%) as possible without the risk of significant hypoglycemia.[22] Patients were allowed to be taking any medications that may affect glycemic control. We excluded patients with gestational diabetes mellitus and those who were unable to attend the group sessions for reasons such as lack of transportation, psychiatric instability (actively suicidal, psychotic), or organic brain injury that precluded them from diabetes self-care.

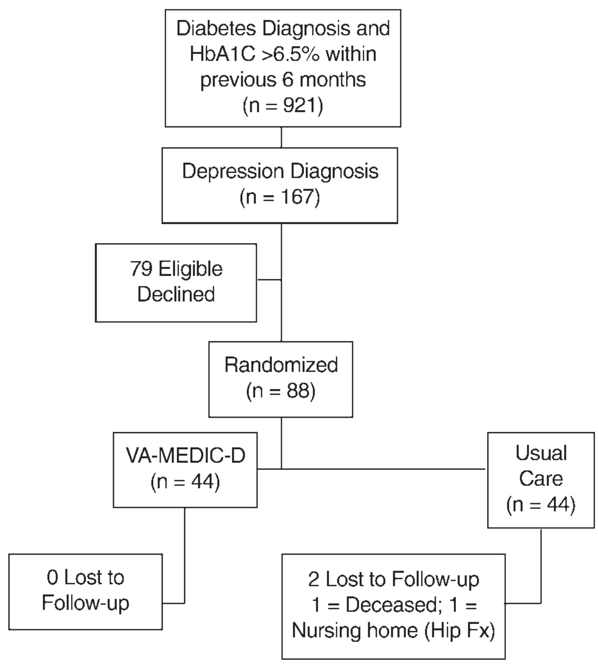

Figure 1. Enrollment, randomization, and follow-up of study participants. A1C = hemoglobin A_{1c}; VA-MEDIC-C = Veterans Affairs Multidisciplinary Educaiton in Diabetes and Intervention for Cardiac Risk Reduction in Depression.

VA-MEDIC-D INTERVENTION

In addition to attending regular visits with a primary care provider, participants in the VA-MEDIC-D intervention arm attended 4 once-weekly sessions of 2 hours, followed by 5 monthly booster sessions held in a classroom with approximately 4-6 participants in each session. Family members, friends, or other sources of social support were encouraged to participate in the sessions. During the informed consent process and at the beginning of session 1, participants and social support members were advised that any information obtained from them by study investigators would remain confidential and safeguarded in accordance with the Privacy Act of 1974; however, any information that the participants or support members disclosed in group might not be kept confidential by the other participants.

Each session consisted of 2 parts: education in the first half, and behavioral and pharmacologic interventions for hypertension, hyperlipidemia, and hyperglycemia and tobacco use in the second half. The education portion was uniform among the sessions and within each discipline, since we standardized and shared handouts and presentation slides. Providers were also required to observe group sessions prior to participation to maintain consistency. No specific intervention was performed for the treatment of depression.

The education portion usually lasted 40-60 minutes and included interactive lectures provided by a nurse, nutritionist, or the clinical pharmacists who were experts on the curriculum of the ADA and certified in the area of diabetes education.[23] Each session focused on 1 or 2 self-care behaviors, such as goal setting, to promote health and problem solving for daily living or integration of psychosocial adjustment to daily life (Appendix II). For example, the second session focused on establishing goals for healthy eating. During this session, the benefits of healthy eating were discussed. Participants reviewed their personal dietary log and identified ways to modify their dietary intake that were consistent with ADA recommendations.[23] At each session, food logs were reviewed by the pharmacist and participants were reminded of their nutrition goals (ie, change a high-fat snack choice to a low-fat choice) to attempt to achieve nutritional goals as suggested by the nutritionist or clinical pharmacist. Participants also prepared healthy food choices during these sessions and were advised of the availability of nutrition programs.

The pharmacologic and behavioral intervention portion lasted 60-80 minutes and was conducted by a clinical pharmacist who had at least 1 year of ambulatory care clinical/training experience and was certified in diabetes education at either the state or national level. The clinical pharmacist would perform a group assessment to determine the degree to which patients felt they could manage the daily aspects of diabetes care (ie, visual acuity to use a glucometer, ability to prepare meals) through a discussion and the Perceived Competence for Diabetes Scale on which patients rated their level of confidence for a particular self-care measure based on a 4-item 7-point Likert scale.[26] The clinical pharmacist was then able to enhance self-efficacy in performing self-care behaviors through the use of group counseling and reinforcement.

Each participant was provided with an individualized cardiovascular risk report card that contained medical history, current medications, vital signs, and laboratory test values obtained prior to the sessions. These report cards were reviewed with patients during the week 1 session and reissued after any pertinent laboratory measurements were determined throughout the study, such as a repeat lipid panel or A1C. Based on these report cards, medications for blood pressure, cholesterol, diabetes, and tobacco cessation were initiated or titrated based on previously established treatment algorithms that were consistent with the VA hospitals' national formulary.[27] No changes were made by the clinical pharmacists for psychiatric medications. Each group member was provided with individualized homework for medication changes and a behavior change goal, such as exercise recommendations, dietary modifications, and blood glucose or blood pressure monitoring. Behavioral interventions were delivered by the clinical pharmacist using constructs from the social cognitive theory,[28-30] which posits that people gain knowledge through observation of others and perceived self-efficacy. The clinical pharmacist used counseling and reinforcement in the group setting to change outcome expectations and expectancies to increase behaviors that would improve diabetes self-care behaviors such as increasing physical activity and healthy eating. Demonstration and coaching to increase self-efficacy for self-care skills, such as monitoring of blood glucose and logging daily dietary intake, were also performed.[28] The number of behavioral change goals was negotiated and based on each patient's competence in the ability to perform these self-care measures.

For tobacco cessation, the behavioral interventions performed by the clinical pharmacist were based on the transtheoretical model.[31] The transtheoretical model assesses an individual's readiness to act on a new healthier behavior and provides strategies or processes of change to guide the individual through the stages of change to action and maintenance. Intervention strategies were tailored to each participant's current stage of change. For example, smokers in the preparation stage were encouraged to cut their cigarette use by half and were provided with handouts with information about treatment options.

STANDARD CARE

The standard of care for patients with diabetes in the Providence VAMC is provided through individual clinic visits with a primary care provider of approximately 30 minutes. All participants randomized to standard care were provided a referral to the VAMC Diabetes Self-Management Education Program (DSME). The DSME program consists of 4 once-weekly educational visits of 2 hours' duration followed by monthly 90-minute follow-up appointments provided by pharmacists, nutritionists, and nurses who were nationally certified diabetes educators. The DSME at the VA uses the ADA curriculum, which covers learning objectives similar to those of the VA-MEDIC-D intervention.[23] Providers who teach these sessions do not adjust medication therapies, but may refer the patient to his or her primary care provider when necessary. Both VA-MEDIC-D and standard care participants were allowed to continue their standard care visits with their mental health providers.

OUTCOMES

The primary outcome was the change in the proportion of participants who attained a goal A1C of <7% at 6 months. Additional outcomes of interest were the proportion of participants who attained the ADA guidelines recommendations for blood pressure and fasting lipid levels, as well as the absolute change in these values.[22] Adherence to ADA guidelines was defined as systolic blood pressure (SBP) <130 mm Hg, diastolic blood pressure (DBP) <80 mm Hg, total cholesterol <200 mg/dL, low-density lipoprotein cholesterol (LDL-C) <100 mg/dL, high-density lipoprotein cholesterol (HDL-C) <130 mg/dL, and tobacco cessation for VA-MEDIC-D versus standard care. It was clinically important to evaluate the absolute change in these values, since studies have demonstrated that a 0.9% reduction in A1C has correlated with a 25% reduction in microvascular complications and a 10-mm Hg reduction in SBP has correlated with a 12% reduction in any diabetes-related complication without a threshold in risk for these parameters.[32,33] All A1C tests were performed at Providence VAMC using liquid chromatography, with a coefficient of variation (CV) of 1.1% within control data (accurate results are considered to be <3% CV); therefore, we can expect the true A1C value to lay within a standard deviation of ±0.1% for an A1C between 5.0% and 10.0%. We also compared the change in the 10-year coronary event risk at 6 months compared to baseline as measured by the United Kingdom Prospective Diabetes Study (UKPDS) risk engine.[34] The UKPDS risk engine provides a single validated comprehensive estimate of the 10-year risk of cardiac events and includes age at diabetes diagnosis, sex, race, current tobacco smoking status, A1C, SBP, and total cholesterol to HDL-C ratio.[34,35] Since peer support, improvement of diabetes measures, and overall health can potentially improve depression symptoms, we also evaluated the change from baseline in the depression symptoms as assessed by the Patient Health Questionnaire-9 (PHQ-9), even when pharmacologic treatment for depression symptoms was not offered by the VA-MEDIC-D providers. The PHQ-9 is a 9-item validated diagnostic instrument that is a reliable measure of depression severity and depression symptom burden and is responsive to treatment outcomes.[36,37] Scores can range from 0 to 27, with a higher score indicating a greater burden of depressive symptomatology.

DATA COLLECTION

Demographic variables, active medication use, and comorbid health conditions (coronary artery disease, hypertension, hypercholesterolemia, anxiety, bipolar disorder, schizophrenia, or posttraumatic stress disorder) were obtained from the electronic medical record at baseline. Laboratory values, blood pressure, weight, and depressive symptoms were obtained at baseline and 6-month follow-up visits for all study participants. Perceived competence for performing diabetes self-care behaviors was assessed using the 4-item Perceived Competence for Diabetes Scale (PCDS), and adherence to diabetes self-care behaviors was tracked with the Summary of Diabetes Self-Care Activities (SDSCA) questionnaire at baseline and 6-month follow-up for all study participants.[26,38] Emergency department (ED) care, hospitalization, and death were defined as serious adverse events and tracked for all study participants.

POWER ANALYSIS

The original sample size of 90 patients (45 in each arm) provided 70% power to detect a 25% difference in change in the proportion that attained an A1C of 7% with the use of a 2-tailed test with an α value of 0.05.

STATISTICAL ANALYSIS

We performed an intent-to-treat analysis. Baseline characteristics were compared between the treatment and control arms using Student *t*-tests for continuous variables or likelihood ratio χ^2 tests for discrete variables. We compared the change in the percentage of participants who attained an A1C <7% at 6 months using stepwise logistic regression, adjusting for their baseline A1C values; baseline PHQ-9 depression scores; total number of primary care, specialty care (ie, cardiology, nephrology, ophthalmology/optometry, and podiatry), mental health, diabetes self-management education, nutrition, individual pharmacist and VA-MEDIC-D visits; diagnosis of schizophrenia; schizoaffective disorder, bipolar disorder, anxiety, and personality disorders; as well as intensification of antidepressant treatment over the study period. We also compared the change from baseline in A1C values at 6 months using a linear regression model to adjust for the baseline values. For missing data of patients lost to follow-up (n = 2 from the standard care arm), we imputed the mean value of the standard care arm for the respective value. We also conducted a sensitivity analysis that included only patients with complete follow-up data (without imputed values). We compared change from baseline in process measures such as PHQ-9 scores, perceived competence, and diabetes self-care activities between the study arms. All statistical analyses were conducted using Stata SE 11.0 (StataCorp, College Station, TX). Statistical significance was defined as $p < 0.05$. Data are presented as mean (SD).

Results

Approximately 1 of 2 eligible patients who were approached agreed to participate (Figure 1). Of the 88 patients randomized, 2 patients from the standard care arm were lost to follow-up. Participants in the VA-MEDIC-D intervention arm attended a mean of 6.6 (2.4) SMA visits and 1.2 (0.8) visits with their primary care provider. Standard care participants attended a mean of 1.3 (0.9) primary care provider visits and 0.6 (1.2) DSMEs over the 6-month study period. VA-MEDIC-D participants had a lower level of guideline adherence for SBP than standard care at baseline (Table 1), but were similar in other cardiovascular guideline adherence rates, cardiovascular risk factor values, and psychiatric comorbidity. At baseline, 11.4% of VA-MEDIC-D versus 25.0% of standard care patients ($p = 0.10$) had attained an A1C <7%, but >6.5%.

After adjusting for baseline A1C values, the magnitude of change after 6 months in the proportion of participants in the VA-MEDIC-D arm achieving guideline adherence for A1C was significantly greater than in the standard care arm (29.6% vs 11.9%), with OR 3.3 (95% CI 1.0 to 10.0). However, the changes in guideline adherence rates for SBP, LDL-C, non–HDL-C, and tobacco use were not significantly different (Figure 2). VA-MEDIC-D participants also achieved significant reductions in absolute values for SBP, LDL-C, non–HDL-C, and A1C from baseline, whereas only significant reductions in non–HDL-C were attained in the standard care arm (Table 2); however, the difference in the degree of change was not significant between the 2 groups. In an adjusted stepwise regression analysis, participation in the VA-MEDIC-D arm was associated with higher A1C goal attainment (adjusted OR 4.3; 95% CI 1.2 to 15.9). Our sensitivity analysis that included participants with complete follow-up data yielded results similar to those presented (data not shown). There was no significant change in the number of patients who used tobacco in the VA-MEDIC-D or standard care groups from baseline (tobacco cessation rate, 1 of 12 smokers vs 2 of 11 smokers; $p = 0.56$, respectively). Of those who continued to use tobacco, 3 of 12 VA-MEDIC-D smokers and 3 of 11 standard care patients had reduced their average daily number of cigarettes by half from baseline ($p = 0.90$). The mean number of cigarettes decreased from 14.3 (10.2) to 11.9 (9.3) for VA MEDIC- D participants ($p = 0.26$) and increased from 12.5 (9.3) to 17.2 (13.6) for standard care participants ($p = 0.22$).

When the overall cardiovascular risk profile was taken into account, the change in the 10-year UKPDS risk decreased significantly for VA-MEDIC-D participants, whereas there was no significant change in the standard care participants, but the magnitude of change did not differ significantly between treatment arms after adjusting for baseline UKPDS values.

Table 1. Comparison of Baseline Characteristics Between VA-MEDIC-D and Standard Care

Characteristic	VA-MEDIC-D	Standard Care	p Value
Participants, n	44	44	
Age, y, mean (SD)	60.2 (9.3)	61.4 (9.9)	0.56
Duration of diabetes diagnosis, y, mean (SD)	9.5 (10.1)	9.3 (7.4)	0.90
Type 1 diabetes, %	0	0	1.00
Type 2 diabetes, %	100	100	1.00
Male, %	100	95.5	0.15
White, %	97.7	100	0.31
Coronary artery disease, %	50.0	47.7	0.83
Tobacco use, %	29.3	25.6	0.70
PHQ-9 score, mean (SD)	10.9 (7.4)	10.5 (7.4)	0.77
Anxiety, %	30.2	37.2	0.49
Posttraumatic stress disorder, %	37.2	37.2	1.00
Bipolar disorder, %	14.0	14.0	1.00
Schizophrenia/schizoaffective disorder, %	0	4.7	0.15
Personality disorder, %	2.3	7.0	0.31
10-y UKPDS risk score, %, mean (SD)	20.6 (14.8)	22.7 (17.8)	0.55
SBP, mm Hg, mean (SD)	130.6 (21.9)	125.2 (16.5)	0.20
DBP, mm Hg, mean (SD)	74.2 (8.2)	72.0 (10.7)	0.27
Non–HDL-C, mg/dL, mean (SD)	137.8 (45.7)	149.6 (50.0)	0.25
LDL-C, mg/dL, mean (SD)	101.0 (29.9)	101.5 (35.3)	0.95
A1C, %, mean (SD)	8.3 (1.7)	8.5 (1.9)	0.70
BMI, kg/m^2, mean (SD)	35.4 (8.5)	33.2 (5.9)	0.16
SBP <130 mm Hg, %	45.5	67.4	0.04
DBP <80 mm Hg, %	81.8	72.1	0.28
Non–HDL-C <130 mg/dL, %	43.2	47.7	0.67
LDL-C <100 mg/dL, %	54.6	61.4	0.52
A1C <7% ,%	11.4	25.0	0.10

A1C = hemoglobin A_{1c}; BMI = body mass index; DBP = diastolic blood pressure; HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol; PHQ-9 = Patient Health Questionnaire-9; SBP = systolic blood pressure; UKPDS = United Kingdom Prospective Diabetes Study; VA-MEDIC-D = Veterans Affairs Multidisciplinary Education in Diabetes and Intervention for Cardiac Risk Reduction in Depression.

VA-MEDIC-D and standard care participants had similar changes in PCDS and SDSCA scores from baseline (Table 3). The mean change in PHQ-9 score was not significantly different for VA-MEDIC-D participants (–3.1 [7.2]; range –21 to 17) vs standard care participants (–1.8 [8.0]; range –24 to 13; p = 0.43), where negative values indicate improvement and positive values indicate worsening of depressive symptoms. PHQ-9 scores decreased by 50% from baseline for 45.5% of VA-MEDIC-D participants and 34.1% of standard care participants (p = 0.28).

MEDICATION CHANGES

VA-MEDIC-D participants were more likely than those in the standard care arm to have a dose increase or initiation of any antihypertensive (68.2% vs 34.9%, p = 0.003), and any antihyperglycemic agent (81.8% vs 58.1%, p = 0.02). There was no significant difference in the addition or dose titration of any antihyperlipidemic between the VA-MEDIC-D and standard care arms (52.3% vs 39.5%, respectively; p = 0.28). There was also no significant difference in the initiation or dose titration of antidepressants for VA-MEDIC-D and standard care participants (29.6% vs 25.0%, respectively; p = 0.81) (Table 4).

HOSPITALIZATIONS, EMERGENCY DEPARTMENT VISITS, AND DEATHS

The mean number of interim care visits with a primary care provider during the study was similar for the VA-MEDIC-D and standard care participants (0.3 [0.6] vs 0.3 [0.6], respectively; p = 0.7). Use of ED services for VA-MEDIC-D and standard care participants did not differ for all-cause visits (39.5% vs 39.5%; p = 1.0), and did not differ significantly for diabetes-related ED visits (2.3% vs 4.7%, respectively; p = 0.6). There was no significant difference in hospital admission rates for VA-MEDIC-D and standard care participants (14.0% vs 16.3%, respectively; p = 0.8). There were no diabetes-related admissions or deaths for either group during the study.

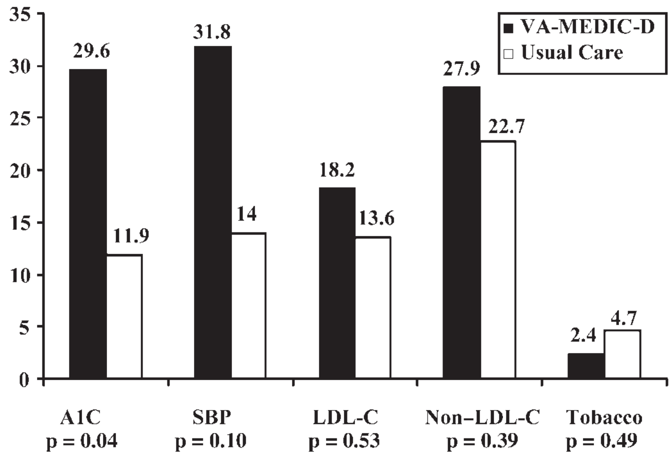

Figure 2. Change in proportion of patients attaining American Diabetes Association guideline recommendations for cardiovascular risk factors. Target goals: hemoglobin A_{1c} (A1C) <7%; systolic blood pressure (SBP) <130 mm Hg; low-density lipoprotein cholesterol (LDL-C) <100 mg/dL; non–high-density lipoprotein cholesterol (HDL-C) <130 mg/dL. Comparison adjusted for baseline values.

Discussion

Our pharmacist-led SMA, VA-MEDIC-D, achieved a greater change in attainment of glycemic control compared with standard care and showed significant reductions from baseline in lipid levels and blood pressure in patients with diabetes and depression, without an increase in serious adverse events. These findings are im-

Table 2. Comparison Between VA-MEDIC-D and Standard Care Follow-up Values and Absolute Change from Baseline in Cardiovascular Risk Factors

	VA-MEDIC-D, mean (SD) (n = 44)		Standard Care, mean (SD) (n = 44)	
Parameter	Follow-up	Absolute Change	Follow-up	Absolute Change
A1C, %	7.4 (1.2)[a]	–0.9 (1.6)[a,b]	8.4 (2.0)	0.0 (1.8)
SBP, mm Hg	123.4 (12.3)	–7.1 (21.5)[b]	127.0 (17.3)	1.8 (18.8)
LDL-C, mg/dL	92.5 (24.3)	–8.6 (26.7)[b]	93.9 (30.6)	–7.6 (35.1)
Non–HDL-C, mg/dL	126.6 (35.5)	–11.2 (34.5)[b]	133.9 (37.0)	–15.7 (49.1)[b]
10-Year UKPDS risk score, %[34]	15.7 (13.1)[b]	–4.9 (8.9)	20.4 (14.6)	–2.3 (15.2)

A1C = hemoglobin A_{1c}; HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol; SBP = systolic blood pressure; UKPDS = United Kingdom Prospective Diabetes Study; VA-MEDIC-D = Veterans Affairs Multidisciplinary Education in Diabetes and Intervention for Cardiac Risk Reduction in Depression.
[a]p < 0.05 VA-MEDIC vs standard care.
[b]p < 0.05 from baseline.

portant because, to our knowledge, our study represents the first of its kind to show an improvement in glycemic control through SMAs in a population with depression.[11,39,40]

Adherence to health care visits is associated with good glycemic control in diabetes.[2] However, the impact of individual visits may be limited in depressed patients because they do not address the underlying components of nonadherence that compromise diabetes care, such as the lack of motivation and social support. The SMA model provides the social dynamic to improve motivation and disease coping through peer support and provides the environment in which the clinical pharmacist is able to efficiently promote behavior change by enhancing conviction and confidence through the use of group counseling and vicarious reinforcement.[28] However, data to support SMAs for the treatment of diabetes are conflicting, and no data previously existed to support SMAs when participants have complex chronic illnesses such as diabetes and depression.

SMAs for diabetes have been led by physicians and nurses, with variable results.[19,20,41] Our previous study showed that pharmacist-led SMAs are feasible and efficacious in improving cardiovascular risk factors in diabetes.[24] Pharmacist-led intervention models may be less costly than physician-led interventions when considering a pharmacist's salary.[42] However, since the efficacy of a pharmacist-led SMA in diabetes and depression was unknown, we did not formally track costs or present a cost-identification analysis. Since our current study provides further support for the use of clinical pharmacists in non–physician-based SMA programs, we intend to track cost in a large-scale randomized controlled trial. This information will be important in light of the current political climate and the desire for highest quality, lowest cost health care.

Our positive findings in glycemic control were achieved despite the lack of a statistically significant impact on depression symptoms. However, VA-MEDIC-D patients experienced a trend toward reduction of depressive symptoms. Previous research has shown that concomitant depression attenuates the impact of traditional interventions and failed to show improvement of glycemic outcomes when interventions were focused on depression treatment.[2,10,11,40] Although not significant, our study showed a trend toward greater improvement for VA-MEDIC-D participants in adherence to self-care behaviors and perceived competence scores. The improvement in glycemic control may have occurred as the result of aggressive medication titration in combination with improvements in adherence and self-perceived competence in performing self-care behaviors as captured by the SDSCA and PCDS scales, respectively, through behavior modeling, which may have not significantly changed the depressive symptoms.[26,38] The question remains as to whether a concomitant treatment for depression would have further improved our gains in glycemic control.

Our study has limitations. First, it was performed at a single site with a relatively homogenous population of white, male veterans with a similar level of social support and psychiatric comorbidities, and it is unknown whether our experience can be reproduced in other health care settings and populations. Second, given our small number of participants, we were underpowered to evaluate the efficacy of our intervention at the subgroup level, such as in patients

Table 3. Change in Depressive Symptoms, Diabetes Self-Care Activities, and Self-Perceived Competence

	VA-MEDIC-D Score, mean (SD) (n = 44)		Standard Care Score, mean (SD) (n = 42)	
Scale	Baseline	Follow-up	Baseline	Follow-up
PHQ-9[a]	10.9 (7.4)	7.8 (6.9)	10.5 (7.4)	8.6 (7.7)
SDSCA	18.7 (6.7)	22.4 (6.1)[b]	17.2 (7.9)	20.1 (7.6)[b]
PCDS	21.0 (0.9)	22.7 (0.8)	21.1 (0.9)	22.0 (0.9)

PCDS = Perceived Competence Diabetes Scale; PHQ-9 = Patient Health Questionnaire-9 Depression Scale; SDSCA = Summary of Diabetes Self-Care Activities.

[a]A PHQ-9 score of 5-9 indicates minimal symptoms; 10-14 is mild major depression, minor depression, or dysthymia; 15-19 indicates moderately severe major depression; and >20 indicates severe major depression.

[b]$p < 0.05$ from baseline.

Table 4. Percentage of Medications Added or Dosages Increased

	VA-MEDIC-D, n = 44		Standard Care, n = 42	
Drug	% Drug Added	% Dose Increased	% Drug Added	% Dose Increased
Aspirin	11.4	13.6	4.6	9.3
ACE inhibitor/ARB	9.1	43.2	6.8	23.3
β-Blocker	6.8	18.2	7.0	11.4
Diuretic	25.0	4.5	13.7	0
Calcium channel blocker	4.5	4.5	2.3	2.3
Sulfonylurea	9.1	9.1	2.3	11.6
Metformin	6.8	18.2	7.0	9.1
Thiazolidinedione	4.6	11.4	2.3	2.3
Insulin	13.6	52.3	14.0	41.9
Statin	18.2	36.4	9.3	25.0
Niacin/fibrate	15.9	22.7	14.0	18.6
Antidepressant	13.6	20.5	13.6	15.9
Anxiolytic	6.8	2.3	11.4	2.3
Antipsychotic	0	4.6	4.6	2.3
Mood stabilizer	11.4	4.6	6.8	6.8

ACE = angiotensin converting enzyme; ARB = angiotensin receptor blocker.

with concomitant schizophrenia or bipolar disorder as well as in different degrees of depression. Third, only 50% of the people who were eligible agreed to participate in our study, which may limit the generalizability of our findings. This finding highlights that convenience of scheduling may be important to patients and further progress should be made to determine the needs of a variety of patients. Finally, we did not present a cost assessment of the VA-MEDIC-D intervention, as efficacy has not been proven.

This study suggests that a pharmacist-led SMA model added to standard care is feasible and more efficacious than standard care alone in attaining glycemic control in patients with diabetes and comorbid depression. A multisite study with formal cost identification analysis is in the planning phase to assess the cost-effectiveness of this intervention.

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Appendix I. ICD-9 Codes

ICD-9 Code	Definition
311	Depressive disorder, not elsewhere classified
	depressive disorder not otherwise specified
	depressive state not otherwise specified
	depression not otherwise specified
296.2	Major depressive episode, single episode
	depressive psychosis, single episode or unspecified
	endogenous depression, single episode or unspecified
	involutional melancholia, single episode or unspecified
	manic-depressive psychosis or reaction, depressed type, single episode or unspecified
	monopolar depression, single episode or unspecified
	psychotic depression, single episode or unspecified
296.3	Major depressive episode, recurrent episode
	any condition classifiable to 296.2, stated to be recurrent

ICD-9 = *International Classification of Diseases*. 9th ed.

Appendix II. Multidisciplinary Education in Diabetes and Intervention for Cardiovascular Risk Reduction in Depression: Program Overview

Session	Part I: Education[a] Learning Objective Focus	Part II: Intervention[b]
Week 1	Diabetes/depression disease process Acute complications	Goal setting Use of self-monitoring equipment Diabetes/cardiovascular risk report card Pharmacologic and behavioral management of diabetes and cardiovascular risk factors Homework
Week 2	Incorporating healthy eating	Review of glucose, activity dietary logs, and report card
Week 3	Integrating physical activity	Goal setting
Week 4	Promoting psychosocial adjustment	Pharmacologic case management
Month 1	Preventing, detecting, and treating chronic complications	Behavioral counseling
Month 2	Goal setting to promote health	Homework
Month 3	Problem solving for daily living	
Month 4	Healthy eating cooking demonstration	
Month 5	Exposure to community support systems	

[a]Delivered by registered nurse, registered dietitian, or PharmD.
[b]Delivered by PharmD.

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Citas Médicas Grupales Conducidas por Farmacéuticos para el Manejo de Diabetes Tipo 2 con Depresión Co-Mórbida en Adultos Mayores

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EXTRACTO

TRASFONDO: La depresión está asociada con un pobre control glicémico, con un número mayor de complicaciones micro y macro vasculares, impedimentos funcionales, mortalidad, y costos de cuidado de salud totales 4.5 veces más altos, en pacientes con diabetes. Las citas médicas compartidas (SMA) pueden ser un método efectivo para lograr recomendaciones de pautas nacionales para el control glicémico en diabetes para aquellos con depresión, a través de apoyo paritario, asesoramiento, solución de problemas, y mejor acceso al cuidado

OBJETIVO: Probar la eficacia de las visitas de grupos SMA dirigidas por farmacéuticos, "Educación Multidisciplinaria en Diabetes e Intervención para la Reducción del Riesgo Cardíaco en Depresión del VA" (VA-MEDIC-D), en pacientes con diabetes tipo 2, según evaluada por la obtención de una hemoglobina A_{1c} (A1C) <7%.

MÉTODOS: Una prueba controlada, aleatoria de VA-MEDIC-D en adición al cuidado estándar contra el cuidado estándar solo, en pacientes deprimidos con diabetes, con una hemoglobina A1C >6.5%. El VA-MEDIC-D consistió de 4 sesiones semanales de 2 horas seguidas por 5 sesiones grupales mensuales de 90 minutos. Cada sesión SMA consistió de educación multidisciplinaria e intervenciones de comportamiento y farmacológicas para la diabetes, los lípidos, el fumar, y la presión sanguínea, dirigidas por farmacéuticos. No se proveyeron intervenciones farmacológicas para la depresión. El cambio en la proporción de participantes que alcanzó un A1C <7% a los 6 meses fue comparado.

RESULTADOS: Comparado con el cuidado estándar (n = 44), el VA-MEDIC-D (n = 44) tuvo una proporción más baja de presiones sanguíneas sistólicas (SBP) <130 mm Hg al comienzo; pero fueron similares en otros factores de riesgo cardiovasculares y co-morbilidades siquiátricas. El cambio en la proporción de participantes que alcanzó un A1C <7% fue mayor en VA-MEDIC-D que en el brazo del cuidado estándar (29.6% vs 11.9%), proporción de probabilidad: 3.6 (95% CI 1.1-12.3). Los participantes en VA-MEDIC-D también lograron reducciones significativas en SBP, colesterol LDL y colesterol no-HDL de base, mientras que el cuidado estándar sólo alcanzó reducciones significativas en el colesterol no-HDL. No hubo cambio significativo en los síntomas de depresión en ninguno de los brazos del estudio

CONCLUSIÓNES: Las visitas de grupos SMA dirigidas por farmacéuticos son eficaces en alcanzar un control glicémico en pacientes con diabetes y depresión, sin cambio en los síntomas de depresión.

Traducido por Brenda R Morand

Rendez-vous Médicaux de Groupe Dirigés par des Pharmaciens Pour la Gestion du Diabète Type 2 Associé à la Dépression.

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RÉSUMÉ

INTRODUCTION: La présence de dépression chez le patient diabétique est associée à un contrôle sous-optimal de la glycémie, un nombre plus élevé de complications micro et macro vasculaires, une déficience fonctionnelle, une mortalité supérieure, et des coûts de santé 4.5 fois plus élevés. Les rendez-vous médicaux de groupe (RVMG) peuvent représenter une façon efficace d'atteindre les directives nationales en matière de contrôle de la glycémie chez les individus diabétiques souffrant de dépression. Ces RVMG mettent à profit les pairs pour le soutien, les conseils et la résolution de problème, tout en améliorant l'accès aux soins.

OBJECTIF: Évaluer l'efficacité des RVMG dirigées par des pharmaciens à atteindre une proportion d'hémoglobine A_{1c} (AIC) inférieure à 7% chez des patients diabétiques de l'administration des vétérans souffrant aussi de dépression.

DEVIS EXPÉRIMENTAL: Étude contrôlée à double-insu chez des patients de l'administration des vétérans souffrant de diabète et de dépression et ayant une A1C supérieure à 6.5%. Le groupe d'intervention (n = 44) a bénéficié d'un programme multidisciplinaire d'éducation et d'intervention en plus des soins usuels alors que le groupe témoin (n = 44) n'a reçu que les soins usuels. Le programme multidisciplinaire consistait en sessions de groupe dans un premier temps hebdomadaires (4 semaines) et d'une durée de 2 heures suivies dans un deuxième temps de 5 sessions mensuelles de 90 minutes. Chacune des sessions était dirigée par un pharmacien et présentait des interventions pharmacologiques et comportementales portant sur le diabète, les lipides, la cigarette, et la tension artérielle. Aucune intervention pharmacologique portait sur la dépression. Les 2 groupes ont été comparés après 6 mois quant à la proportion de patients ayant une A1C inférieure à 7%.

RÉSULTATS: Sauf pour la tension systolique, inférieure à 130 mm Hg chez une plus grande proportion de patients du groupe d'intervention, les 2 groupes étaient similaires au départ quant aux facteurs de risques cardio-vasculaires et aux comorbidités psychiatriques. Après 6 mois, la proportion de patients ayant atteint une A1C inférieure à 7% était plus élevée dans le groupe d'intervention (29.6% vs11.9%), risque relatif approché: 3.6 (95% IC 1.1-12.3). Les patients du groupe d'intervention ont aussi bénéficié de réductions significatives de la tension systolique, du cholesterol LDL, et du cholesterol non-HDL, alors que seul le cholesterol non-HDL a été significativement réduit chez le groupe témoin. Aucune différence n'a été observée au niveau des symptômes de dépression dans les 2 groupes.

CONCLUSIONS: Des RVMG dirigés par des pharmaciens sont efficaces pour améliorer le contrôle de la glycémie chez les patients diabétiques souffrant aussi de dépression, sans toutefois influencer les symptômes de dépression.

Traduit par Suzanne Laplante